Reg. No:

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccreditedby NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2025

COMMON TO B.E – CSE / B.TECH – IT / AIDS & M.TECH - CSE

20MA214 - DISCRETE STRUCTURES AND COMBINATORICS

Answer ALL questions

Duration: 3 Hours **Maximum: 100 Marks**

PART - A **(7x 1 = 7 Marks)**

1. **CO1** **The simplified form of the Boolean expression** $x.y + x'.z + x.z$ **is _______.**

   a) $x.y + x'.z$ b) $x'.z$ c) $x.y + z$ d) $y + z$

2. **CO2** **A relation** $R = \{(1,3),(2,2),(2,3)\}$ **on a set** $A = \{1,2,3\}$ **is _______.**

   a) Symmetric b) transitive c) reflexive d) equivalence

3. **CO2** **How many relations exist from set X to set Y if the set X and set Y has 2 and 3 elements respectively?**

   a) 6 b) 5 c) 32 d) 64

4. **CO1** **A complemented and distributive lattice is called ______.**

   a) Boolean algebra b) equivalence c) poset d) homomorphism

5. **CO4** **The number of different permutations of the word BANANA is _______.**

   a) 720 b) 60 c) 120 d) 360

6. **CO4** **How many words of three distinct letter can be formed from the alphabet {a,b,y,z}.**

   a) 24 b) 12 c) 36 d) 48

7. **CO4** **Find the value of '** n **' if** $nP_2 = 20$**.**

   a) 1 b) 5 c) 4 d) 2

Fill in the blanks: **(3x 1 = 3 Marks)**

8. **CO1** A partially ordered set is said to be lattice if every pair of elements has _______.

9. **CO3** The contrapositive of $P \rightarrow Q$ is ________.

10. **CO3** A set of premises H_1, H_2,…..H_n is said to be inconsistent, if their conjunction implies a ______.

PART - B **(5x 3 = 15 Marks)**

11. **CO2** If $R = \{(1,2),(2,2),(3,4)\}, S = \{(1,3),(2,5),(3,1),(4,2)\}$ are two relations then find $S \circ R$ and $R \circ S$.

12. **CO2** If A and B are any subsets of Universal set , then prove that $f_{A\cup B}(x) = f_A(x) + f_B(x) - f_{A\cap B}(x)$.

13. **CO3** Without using truth table, prove that $\neg p \rightarrow (q \rightarrow r) \equiv q \rightarrow (p \vee r)$

14. **CO4** How many permutations of the letters A, B, C, D, E, F, G contain
    - the string BCD,
    - the string CFGA

15. **CO1** Draw the Hasse diagram representing the partial ordering {(x, y)/ x divides y} on {2,3,6,12,24,36}.

PART - C **(5 x 15 = 75 Marks)**

Q.No: 16 Compulsory Question
Q.No: 17 to 22 Answer any <u>FOUR</u> Questions

<u>Compulsory Question:</u>

16. **CO1** i) In any Boolean algebra show that $ab' + a'b = 0$ iff $a = b$. (8)

    CO2 ii) Let X=[1,2……7] and R=[(x, y)/x-y is divisible by3}, show that R is an equivalence relation. (7)

17. **CO2** i) If R and S be relations on a set A represented by the matrices (8)

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad M_S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Find the matrices that represent

- $R \cup S$
- $R \cap S$
- $(R \circ S)$
- $S \circ R$
- $R \oplus S$.

	CO2	ii)	Analyze the primitive recursive for the function f(x,y)=x^y	(7)
18.	**CO3**	i)	Without using truth table, find the PCNF and PDNF of $(p \rightarrow (q \wedge r)) \wedge (\neg p \rightarrow (\neg q \wedge \neg r))$	(8)
	CO3	ii)	Show that $\forall x(P(x) \vee Q(x)) \Rightarrow \forall x(P(x) \vee \exists x Q(x)$ using the indirect method.	(7)
19.	**CO4**	i)	Solve the recurrence relation $a_n - 2a_{n-1} = 3^n;\ a_1 = 5$	(8)
	CO4	ii)	Find the number of integers between 1 and 250 both inclusive that are not divisible by any of the integers 2,3,5 and 7.	(7)
20.	**CO3**	i)	Show that the following set of premises is inconsistent: • If Ram gets his degree, he will go for a job. • If he goes for a job, he will get married soon. • If he goes for higher study, he will not get married. • Ram gets his degree and goes for higher study.	(8)
	CO3	ii)	Show that $(t \wedge s)$ can be derived from the premises $p \rightarrow q,\ q \rightarrow \neg r,\ r,$ $p \vee (t \wedge s)$.	(7)
21.	**CO4**	i)	An urn contains 15 balls, eight of which are red and seven are black. In how many ways can five balls be chosen so that • All five are red, • All five are black • Two are red and Three are black • Three are red and two are black.	(8)
	CO3	ii)	Without using truth table prove the following equivalence $p \rightarrow (q \rightarrow p) \equiv \neg p \rightarrow (p \rightarrow q)$	(7)
22.	**CO1**	i)	State and prove Isotonic property in Lattice.	(8)
	CO2	ii)	If $f: Z \rightarrow N$ is defined by $f(x) = \begin{cases} 2x-1, & if\ x > 0 \\ -2x, & if\ x \leq 0 \end{cases}$ • Prove that 'f' is one –to-one and onto. • Determine f^{-1}	(7)

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